

Spring 1987

Problem 1 (Sp87). A standard theorem states that a continuous real valued function on a compact set is bounded. Prove the converse: If K is a subset of $\mathbb{R}^n$ and if every continuous real valued function on K is bounded, then K is compact.

Note: See also ??.

Problem 2 (Sp87). Let the transformation T from $U = \{(u, v) \mid u > v\}$ of $\mathbb{R}^2$ into $\mathbb{R}^2$ be defined by $T(u, v) = (u + v, u^2 + v^2)$.

1. Prove that T is locally one-to-one.
2. Determine the image of T , and show that T is globally one-to-one.

Problem 3 (Sp87). Let f be a complex valued function in the open unit disc, $\mathbb{D}$, of the complex plane such that the functions $g = f^2$ and $h = f^3$ are both analytic. Prove that f is analytic in $\mathbb{D}$.

Problem 4 (Sp87). Let $\mathbb{F}_q$ be a finite field with q elements and let x be an indeterminate. For f a polynomial in $\mathbb{F}_q[x]$, let φ_f denote the corresponding function of $\mathbb{F}_q$ into $\mathbb{F}_q$, defined by $\varphi_f(a) = f(a)$, for $a \in \mathbb{F}_q$. Prove that if φ is any function of $\mathbb{F}_q$ into $\mathbb{F}_q$, then there is an f in $\mathbb{F}_q[x]$ such that $\varphi = \varphi_f$. Prove that f is uniquely determined by φ to within addition of a multiple of $x^q - x$.

Problem 5 (Sp87). Let f be a continuous real valued function on $\mathbb{R}$ satisfying

$$|f(x)| \leq \frac{C}{1 + x^2}$$

where C is a positive constant. Define the function F on $\mathbb{R}$ by

$$F(x) = \sum_{n=-\infty}^{\infty} f(x + n)$$

1. Prove that F is continuous and periodic with period 1.
2. Prove that if G is continuous and periodic with period 1, then

$$\int_0^1 F(x)G(x) \, dx = \int_{-\infty}^{\infty} f(x)G(x) \, dx$$

Problem 6 (Sp87, Fa89). Let f be an analytic function in the open unit disc of the complex plane such that $|f(z)| \leq C/(1 - |z|)$ for all z in the disc, where C is a positive constant. Prove that

$$|f'(z)| \leq \frac{4C}{(1 - |z|)^2}$$

Problem 7 (Sp87). Let p, q and r be continuous real valued functions on $\mathbb{R}$, with $p > 0$. Prove that the differential equation

$$p(t)x''(t) + q(t)x'(t) + r(t)x(t) = 0$$

is equivalent to, i.e., has exactly the same solutions as, a differential equation of the form

$$(a(t)x'(t))' + b(t)x(t) = 0$$

where a is continuously differentiable and b is continuous.

Problem 8 (Sp87). Prove that if the nonconstant polynomial $p(z)$, with complex coefficients, has all of its zeros in the half-plane $\Re z > 0$, then all of the zeros of its derivative are in the same half-plane.

Problem 9 (Sp84, Sp87). Let A be a real $m \times n$ matrix with rational entries and let b be an m -tuple of rational numbers. Assume that the system of equations $Ax = b$ has a solution x in complex n -space $\mathbb{C}^n$. Show that the equation has a solution vector with rational components, or give a counterexample.

Problem 10 (Sp87). Prove that any finite group of order n is isomorphic to a subgroup of $O_n(\mathbb{R})$, the group of $n \times n$ orthogonal real matrices.

Problem 11 (Sp87). Show that the equation $ae^x = 1 + x + \frac{x^2}{2}$, where a is a positive constant, has exactly one real root.

Problem 12 (Sp87). Evaluate the integral $I = \int_0^{1/2} \frac{\sin x}{x} dx$ to an accuracy of two decimal places.

Problem 13 (Fa83, Sp87). Let $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ be a function which is continuously differentiable and whose partial derivatives are uniformly bounded:

$$\left| \frac{\partial f}{\partial x_i}(x_1, \dots, x_n) \right| \leq M$$

for all $(x_1, \dots, x_n) \neq (0, \dots, 0)$. Show that if $n \geq 2$, then f can be extended to a continuous function defined on all of $\mathbb{R}^n$. Show that this is false if $n = 1$ by giving a counterexample.

Problem 14 (Sp87). 1. Show that, to within isomorphism, there is just one noncyclic group G of order 4.

2. Show that the group of automorphisms of G is isomorphic to the permutation group S_3 .

Problem 15 (Sp87). Prove or disprove: If the function f is analytic in the entire complex plane, and if f maps every unbounded sequence to an unbounded sequence, then f is a polynomial.

Problem 16 (Sp87). Let $\mathcal{F}$ be a uniformly bounded, equicontinuous family of real valued functions on the metric space (X, d) . Prove that the function

$$g(x) = \sup\{f(x) \mid f \in \mathcal{F}\}$$

is continuous.

Problem 17 (Sp87). Let V be a finite dimensional linear subspace of $\mathcal{C}_{\mathbb{R}}^{\infty}$, the space of complex valued, infinitely differentiable functions. Assume that V is closed under D , the operator of differentiation, i.e., $f \in V \Rightarrow Df \in V$. Prove that there is a constant coefficient differential operator

$$L = \sum_{k=0}^n a_k D^k$$

such that V consists of all solutions of the differential equation $Lf = 0$.

Problem 18 (Fa83, Sp87, Fa99). Let V be a finite dimensional complex vector space and let A and B be linear operators on V such that $AB = BA$. Prove that if A and B can each be diagonalized, then there is a basis for V which simultaneously diagonalizes A and B .

Problem 19 (Sp87, Fa95). Let $\mathbb{F}$ be a field. Prove that every finite subgroup of the multiplicative group of nonzero elements of $\mathbb{F}$ is cyclic.

Problem 20 (Sp87). Evaluate $I = \int_0^{\pi} \frac{\cos 4\theta}{1 + \cos^2 \theta} d\theta$.